

Test Form B

Name _____ Date _____

Chapter 11

Class _____ Section _____

1. Find a unit vector in the direction of $\mathbf{v}$ if $\mathbf{v}$ is the vector from $P(2, -1, 3)$ to $Q(1, 0, -4)$.

(a) $\mathbf{i} - \mathbf{j} + 7\mathbf{k}$

(b) $-\mathbf{i} + \mathbf{j} - \mathbf{k}$

(c) $-\frac{1}{\sqrt{51}}\mathbf{i} + \frac{1}{\sqrt{51}}\mathbf{j} - \frac{7}{\sqrt{51}}\mathbf{k}$

(d) $\frac{1}{\sqrt{51}}\mathbf{i} - \frac{1}{\sqrt{51}}\mathbf{j} + \frac{1}{\sqrt{51}}\mathbf{k}$

(e) None of these

2. Calculate $\mathbf{j} \times \mathbf{k}$.

(a) $-\mathbf{i}$

(b) $-\mathbf{k}$

(c) $-\mathbf{j}$

(d) $\mathbf{i}$

(e) None of these

In problems 3 through 8, let $\mathbf{u} = \mathbf{i} + 2\mathbf{j} - \mathbf{k}$, $\mathbf{v} = 3\mathbf{i} - \mathbf{j} + 4\mathbf{k}$ and $\mathbf{w} = 2\mathbf{j} - \mathbf{k}$.3. Calculate $\mathbf{u} \cdot \mathbf{v}$.

(a) $\langle 7, -7, -7 \rangle$

(b) -3

(c) $\langle -1, 1, 1 \rangle$

(d) 24

(e) None of these

4. Calculate $\mathbf{u} \times \mathbf{v}$.

(a) $\langle 7, -7, -7 \rangle$

(b) -3

(c) $\langle -1, 1, 1 \rangle$

(d) 24

(e) None of these

5. Calculate $(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{w}$.

(a) -4

(b) 0

(c) -7

(d) $\mathbf{i} - 8\mathbf{j}$

(e) None of these

6. Calculate the component of $\mathbf{u}$ in the direction of $\mathbf{v}$.

(a) $\frac{24}{\sqrt{26}}$

(b) $-\frac{3}{\sqrt{26}}$

(c) $\frac{9}{\sqrt{26}}$

(d) $-\frac{9}{26}\mathbf{i} + \frac{3}{26}\mathbf{j} - \frac{6}{13}\mathbf{k}$

(e) None of these

7. Calculate $\cos \theta$ where θ is the angle between $\mathbf{u}$ and $\mathbf{v}$.

(a) $\frac{9}{2\sqrt{26}}$

(b) $\frac{9}{2\sqrt{39}}$

(c) $\frac{-3}{2\sqrt{26}}$

(d) $\frac{-3}{2\sqrt{39}}$

(e) None of these

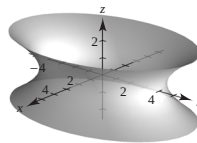
8. Which of the following vectors is orthogonal to both $\mathbf{u}$ and $\mathbf{w}$?
- (a) $\mathbf{j}$ (b) $\mathbf{w} \times \mathbf{u}$ (c) $\mathbf{v}$
 (d) $4\mathbf{j} + \mathbf{k}$ (e) None of these
9. Which of the following statements is true about the vectors $\mathbf{u} = \frac{1}{2}\mathbf{i} + \frac{1}{3}\mathbf{j} - \frac{1}{4}\mathbf{k}$ and $\mathbf{v} = -2\mathbf{i} - \frac{4}{3}\mathbf{j} + \mathbf{k}$?
- (a) $\mathbf{u}$ and $\mathbf{v}$ are orthogonal. (b) $\mathbf{u}$ and $\mathbf{v}$ are parallel.
 (c) $\mathbf{u}$ is a unit vector of $\mathbf{v}$. (d) The angle between $\mathbf{u}$ and $\mathbf{v}$ is $\frac{\pi}{4}$.
 (e) None of these
10. Which of the following is a set of parametric equations for the line through the points $(-3, 2, 0)$ and $(4, 3, 3)$?
- (a) $x = 4 + 7t$
 $y = 3 + t$
 $z = 3 + 3t$ (b) $x = 4 - 3t$
 $y = 3 + 2t$
 $z = 3$ (c) $x = -3 + t$
 $y = 2 + 5t$
 $z = 3t$
 (d) $x = -3 + 4t$
 $y = 2 + 3t$
 $z = 3t$ (e) None of these
11. Find an equation of the plane that passes through the points $(2, 1, -4)$, $(-3, 1, 3)$ and $(-2, -1, 0)$.
- (a) $2x - 4y + 5z + 10 = 0$ (b) $7x + 6y + 5z + 20 = 0$ (c) $7x - 4y + 5z + 10 = 0$
 (d) $7x - 4y + 5z + 2 = 0$ (e) None of these
12. If the equation of a cylinder is given by $x^2 - z = 0$, then
- (a) The rulings are parallel to the x -axis. (b) The graph is a parabola.
 (c) The rulings are parallel to the z -axis. (d) The generating curve is a parabola.
 (e) None of these
13. Calculate the distance from the point $(1, -2, 3)$ to the plane given by $(x - 3) + 2(y + 1) - 4z = 0$.
- (a) $\sqrt{14}$ (b) $\frac{14}{\sqrt{21}}$ (c) $\frac{16}{\sqrt{21}}$
 (d) $\sqrt{65}$ (e) None of these
14. Find the center and radius of the sphere given by the equation $x^2 + y^2 + z^2 - 4x - 2y + 2z = 10$.
- (a) Center: $(-2, -1, 1)$
 Radius: 4 (b) Center: $(2, 1, -1)$
 Radius: 4 (c) Center: $(-2, -1, 1)$
 Radius: $\sqrt{10}$
 (d) Center: $(2, 1, -1)$
 Radius: $\sqrt{10}$ (e) None of these

15. Identify the quadric surface given by $z^2 = \frac{x^2}{4} + \frac{y^2}{16}$.

- (a) Elliptic cone (b) Elliptic paraboloid (c) Hyperbolic paraboloid
(d) Hyperboloid of one sheet (e) Hyperboloid of two sheets

16. Identify the quadric surface sketched at the right.

- (a) Hyperboloid of two sheets (b) Elliptic paraboloid
(c) Hyperboloid of one sheet (d) Elliptic cone
(e) None of these



17. Find the equation of revolution if the generating curve, $y - 2x = 1$, is revolved about the x -axis.

- (a) $4x^2 - y^2 + 4z^2 + 2y - 1 = 0$ (b) $3x^2 - z^2 + 4x + 1 = 0$
(c) $4x^2 - y^2 - z^2 + 4x + 1 = 0$ (d) $3x^2 - y^2 - z^2 + 4x + 1 = 0$
(e) None of these

18. Express the spherical point $\left(3, \frac{3\pi}{4}, \frac{\pi}{3}\right)$ in rectangular coordinates.

- (a) $\left(\frac{-3}{\sqrt{2}}, \frac{3}{\sqrt{2}}, \frac{3\sqrt{3}}{2}\right)$ (b) $\left(\frac{-3\sqrt{3}}{2\sqrt{2}}, \frac{3\sqrt{3}}{2\sqrt{2}}, \frac{3}{2}\right)$ (c) $\left(\frac{3}{2\sqrt{2}}, \frac{3\sqrt{3}}{2\sqrt{2}}, \frac{-3}{\sqrt{2}}\right)$
(d) $\left(\frac{3}{2}, \frac{3\sqrt{3}}{2}, \frac{-3}{\sqrt{2}}\right)$ (e) None of these

19. Find an equation in cylindrical coordinates for the surface given by the rectangular equation $x^2 - y^2 = z$.

- (a) $r^2 = z$ (b) $r^2 = z \sec 2\theta$ (c) $\rho = \cot \phi \csc \phi \sec 2\theta$
(d) $\rho = \cot \phi \csc \phi$ (e) None of these

20. Find the work done in moving a particle from $P(0, 2, 1)$ to $Q(3, -2, 2)$ if the magnitude and direction of the force are given by $\mathbf{v} = \langle 4, -2, 3 \rangle$.

- (a) 15 (b) 21 (c) 168
(d) 23 (e) None of these